Worksheet 3-2 Mixed Derivative Practice (thru 3-2)
AP Calculus

Name SOLUTIONS

Directions: Answer the problems below. Be sure to show all work ON NOTEBOOK PAPER.

1. $f(x) = 4 - 5x - \ln(x+2)$, then $f'(-1) =$

$f'(x) = -5 - \frac{1}{x+2}$ $\quad f'(-1) = -5 - \frac{1}{1} = \boxed{-6}$

2. $g(x) = (x^2 - \sin(x))^4$, then $g'(x) =$

$g'(x) = 4(x^2 - \sin x)^3(2x - \cos x)$

3. $h(x) = 2x^3 - x^2 + \frac{1}{x} - 3$, then $h'(3) =$

$h'(x) = 6x^2 - 2x - \frac{1}{x^2}$ $\quad h'(3) = 6(9) - 6 - \frac{1}{9} = \frac{431}{9}$

4. The function f is defined by $f(x) = \frac{x-1}{x+2}$. Find all points (x, y) where the line tangent to the graph of f has slope $\frac{1}{3}$.

$f'(x) = \frac{(x+2)(1) - (x-1)(1)}{(x+2)^2}$

$= \frac{x+2-x+1}{(x+2)^2}$

$= \frac{3}{(x+2)^2}$

$m_{tan} = \frac{1}{3} \rightarrow f'(x) = \frac{1}{3}$

$\frac{3}{(x+2)^2} = \frac{1}{3}$

$(x+2)^2 = 9$

$x+2 = 3 \quad x+2 = -3$

$x = 1 \quad x = -5$

$f(1) = \frac{1-1}{1+2} = 0 \quad f(-5) = \frac{-5-1}{-5+2} = \frac{-6}{-3} = 2$

$(1,0)$ and $(-5,2)$

5. What is the equation of the line tangent to the graph of $y = \ln(x^2)$ when $x = 3$?

$y(3) = \ln(9)$

$y' = \frac{1}{x^2} \cdot 2x = \frac{2}{x}$

$m_{tan} = y'(3) = \frac{2}{3}$

EQ_{TAN}: $y - \ln 9 = \frac{2}{3}(x-3)$

6. If $f(x) = 4x^{-1} - \frac{x}{3} - 2$, then $f'(-3) =$

$f(x) = \frac{4}{x} - \frac{1}{3}x - 2$

$f'(x) = -\frac{4}{x^2} - \frac{1}{3}$

$f'(-3) = \frac{-4}{9} - \frac{1}{3} = \frac{-7}{9}$

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x	-6	-2	6
$f(x)$	3	-4	5
$f'(x)$	2	1	-3

7. The table above gives selected values for the differentiable function f and its derivative f'. Let k be the function defined by $k(x) = \frac{f(3x)}{4x}$. What is the value of $k'(-2)$?

$k'(x) = \frac{4x \cdot f'(3x) \cdot 3 - f(3x) \cdot 4}{(4x)^2}$

$k'(-2) = \frac{(-8)(f'(-6))(3) - (f(-6))(4)}{(-8)^2} = \frac{(-8)(2)(3) - 3(4)}{64} = \frac{-60}{64} = \frac{-15}{16}$

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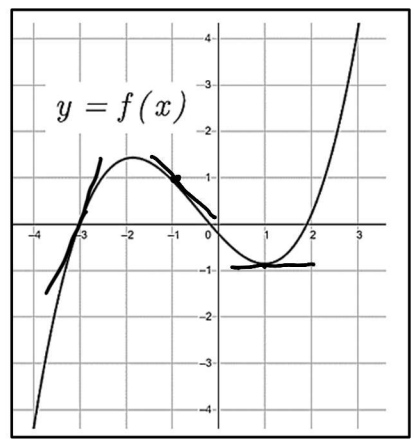
8. The graph of $f(x)$ is below. Order $f'(-3), f'(-1)$, and $f'(1)$ from least to greatest.



$f'(-3) > 0 \quad f'(-1) < 0 \quad f'(1) = 0$

$f'(-1) < f'(1) < f'(-3)$

9. If f is the function given by $f(x) = 4 - x^3$, for which value of x is the instantaneous rate of change equal to the average rate of change of $f(x)$ on the interval $[-1,3]$?

IROC $= f'(x) = -3x^2$

AROC on $[-1,3]$ $= \frac{f(3) - f(-1)}{3 - -1}$

$= \frac{-23 - 5}{4} = \frac{-28}{4} = -7$

$-3x^2 = -7$

$x^2 = \frac{7}{3}$

$x = \pm\sqrt{\frac{7}{3}} \approx \pm 1.5275$

on $[-1,3]$, $x = \sqrt{\frac{7}{3}}$ is the only x-value

10. $\lim_{x\to-\infty} \frac{9x-2}{\sqrt{4x^2-2x+1}} =$

$$= \lim_{x\to-\infty} \frac{9x}{\sqrt{4x^2}} = \lim_{x\to-\infty} \frac{9x}{2|x|} = \frac{9}{2}\cdot\lim_{x\to-\infty}\frac{x}{|x|} = \frac{9}{2}(-1) = -\frac{9}{2}$$

11. $f(x) = (3x^2-5x+1)^6$. Find $f'(x)$

$$f'(x) = 6(3x^2-5x+1)^5(6x-5) = 6(6x-5)(3x^2-5x+1)^5$$

12. $g(x) = \sin^2(3x-1)$. Find $g'(x)$.

$$g(x) = [\sin(3x-1)]^2$$

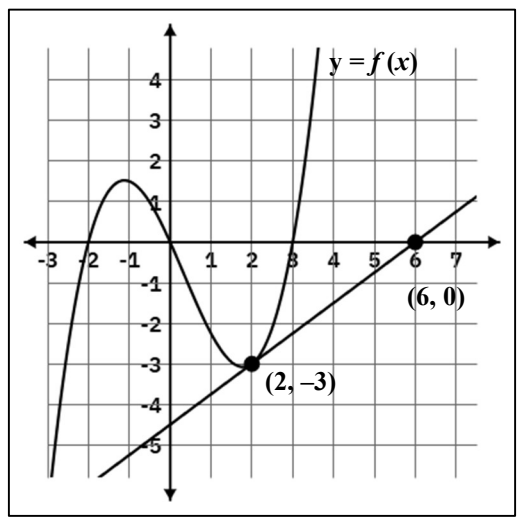
$$g'(x) = 2[\sin(3x-1)]\cdot\cos(3x-1)(3) = 6\sin(3x-1)\cos(3x-1)$$

13. $g(x) = \frac{5-2x}{3x+7}$, find $g'(x)$

$$g'(x) = \frac{(3x+7)(-2)-(5-2x)(3)}{(3x+7)^2}$$

$$= \frac{-6x-14-15+6x}{(3x+7)^2} = \frac{-29}{(3x+7)^2}$$

14. The graph of $f(x)$ below along with the line tangent to f at $x=2$. Let $g(x) = \frac{f(x)}{e^x}$. Find $g'(2)$.

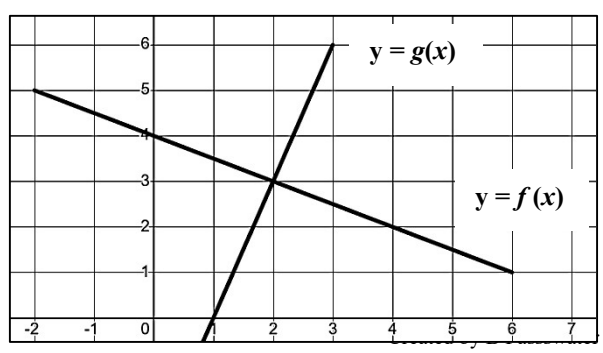


$$g'(x) = \frac{e^x\cdot f'(x) - f(x)\cdot e^x}{(e^x)^2}$$

$$g'(2) = \frac{e^2 f'(2) - f(2)e^2}{e^2\cdot e^2}$$

$$= \frac{f'(2)-f(2)}{e^2} = \frac{\frac{3}{4}-{}^{-}3}{e^2} = \frac{3+12}{4e^2} = \frac{15}{4e^2}$$

15. The figure below shows the graphs of f and g. If $k(x) = f(x)g(x)$, find $k'(2)$.



$$K'(x) = f'(x)g(x) + f(x)g'(x)$$

$$K'(2) = f'(2)g(2) + f(2)g'(2)$$

$$= -\frac{1}{2}(3) + 3(3) = -\frac{3}{2} + 9 = \frac{15}{2}$$

$$g(x) = \begin{cases} ax^2 + 3, & x \le 2 \\ 16x - 23, & x > 2 \end{cases}$$

16. The function g is defined above. Find the value of a that makes g continuous at $x = 2$

$g(2) = a(4) + 3 = 4a + 3$

$\lim_{x \to 2^-} g(x) = \lim_{x \to 2^-} (ax^2 + 3) = 4a + 3$

$\lim_{x \to 2^+} g(x) = \lim_{x \to 2^+} (16x - 23) = 32 - 23 = 9$

g continuous at $x = 2$

$\lim_{x \to 2^-} g(x) = \lim_{x \to 2^+} g(x)$

$4a + 3 = 9$

$4a = 6$

$a = \frac{6}{4} = \frac{3}{2}$

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17. If f is the function given by $f(x) = \sin\left(\frac{x}{6}\right)$, write the equation of the line tangent to the graph of f at the point $\left(\pi, \frac{1}{2}\right)$.

$f'(x) = \cos\left(\frac{x}{6}\right)\left(\frac{1}{6}\right)$

$m_{tan} = f'(\pi) = \frac{1}{6} \cdot \cos\frac{\pi}{6}$

$= \frac{1}{6} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{12}$

EQ_{TAN}: $y - \frac{1}{2} = \frac{\sqrt{3}}{12}(x - \pi)$

18. If $f(x) = \sqrt{x^2 - 4}$ and $g(x) = 3x - 2$, find the derivative of $g(f(x))$ at $x = 6$.

$\frac{d}{dx}[g(f(x))] = g'(f(x)) \cdot f'(x)$

$g'(f(6)) \cdot f'(6) = g'(\sqrt{32}) \cdot f'(6)$

$= 3\left(\frac{6}{\sqrt{32}}\right)$

$= \frac{18}{\sqrt{32}}$

$f'(x) = \frac{1}{2}(x^2 - 4)^{-1/2}(2x)$

$g'(x) = 3$

$f(6) = \sqrt{36 - 4} = \sqrt{32}$

$g'(\sqrt{32}) = 3$

$f'(6) = \frac{1}{2}(32)^{-1/2}(12) = \frac{6}{\sqrt{32}}$

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x	1	2	5	7
$f(x)$	2	8	2	-3

19. Let f be a continuous function with selected values given in the table above. Is there a time c, where $1 < c < 7$, such that $f(c) = 0$? Justify your answer.

Since f is continuous, by the IVT, there is a value c for $1<c<7$, such that $f(7) = -3 < 0 < f(1) = 2$, so there is a value c in $1<c<7$ such that $f(c) = 0$

20. $\lim_{x \to -3} \frac{x^2 - 9}{x^2 - 4x - 21} =$

$$= \lim_{x \to -3} \frac{(x+3)(x-3)}{(x-7)(x+3)} = \lim_{x \to -3} \frac{x-3}{x-7} = \frac{-3-3}{-3-7} = \frac{-6}{-10} = \frac{3}{5}$$

21. Let $3x^2 - 2y + 4xy = 6$. Find $\frac{dy}{dx}$.

$$6x - 2\frac{dy}{dx} + 4y + 4x\frac{dy}{dx} = 0$$

$$-2\frac{dy}{dx} + 4x\frac{dy}{dx} = -6x - 4y$$

$$\frac{dy}{dx}(-2+4x) = -6x - 4y$$

$$\frac{dy}{dx} = \frac{-6x-4y}{-2+4x} = \frac{-3x-2y}{-1+2x}$$

22. Write an equation of the line tangent to the curve $x^2 - 3xy + 4y = 8$ through the point $(2, -2)$.

$$2x - 3y - 3x\frac{dy}{dx} + 4\frac{dy}{dx} = 0$$

$$2(2) - 3(-2) - 3(2)\frac{dy}{dx} + 4\frac{dy}{dx} = 0$$

$$4 + 6 - 6\frac{dy}{dx} + 4\frac{dy}{dx} = 0$$

$$-2\frac{dy}{dx} = -10$$

$$\frac{dy}{dx} = 5$$

EQ TAN:

$$y + 2 = 5(x - 2)$$

23. Let $\sin(2x - 3y) = x^2 + 4y^3$. Find $\frac{dy}{dx}$.

$$\cos(2x-3y)\left[2 - 3\frac{dy}{dx}\right] = 2x + 12y^2\frac{dy}{dx}$$

$$2\cos(2x-3y) - 3\frac{dy}{dx}\cos(2x-3y) = 2x + 12y^2\frac{dy}{dx}$$

$$-3\frac{dy}{dx}\cos(2x-3y) - 12y^2\frac{dy}{dx} = 2x - 2\cos(2x-3y)$$

$$\frac{dy}{dx}\left[-3\cos(2x-3y) - 12y^2\right] = 2x - 2\cos(2x-3y)$$

$$\frac{dy}{dx} = \frac{2x - 2\cos(2x-3y)}{-3\cos(2x-3y) - 12y^2}$$